

1. Construct a truth table for the compound proposition

$$(p \vee q) \wedge (p \vee \neg q) \leftrightarrow p$$

and express this proposition as an English sentence with the propositions "The integer n is even." and "The integer n is greater than 7." for p and q respectively.

2. Determine the negation of the proposition

"For every positive real number $x > 0$, there exists a positive integer n such that $\frac{1}{n} < x$."

and write out both the proposition and its negation using the quantifiers $\exists$ and $\forall$.

3. Show that the proposition

$$p \leftrightarrow q$$

is logically equivalent to the proposition

$$(p \rightarrow q) \wedge (\neg p \rightarrow \neg q).$$

Make up propositions p and q and write this equivalence as an english sentence.

4. Consider the proposition

$$P(x) = \exists y (y^2 \leq x \leq y) \quad \text{for } y \text{ a real number.}$$

Can we determine for which real numbers x the proposition $P(x)$ is true?

Try to find another proposition $Q(x)$ so that $P(x) \equiv Q(x)$ where $Q(x)$ does not contain any quantifiers.